UNIVERSITI TEKNIKAL MALAYSIA MELAKA	**Mini Project** FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY UNIVERSITI TEKNIKAL MALAYSIA MELAKA	
FUZZY LOGIC		
BAXI 3123	**SEMESTER 2**	**SESI 2025/2026**

You are required to complete this project in group. Propose and solve a fuzzy logic problem of your choice based on a topic of your choice. **You need to demonstrate a Fuzzy Systems use case**. You are required to design your solution using **both the knowledge-driven approach** (e.g. Fuzzy Inference System), and the **data driven approach** (e.g. Fuzzy C-Mean or other automated methods) to solve your chosen problem.

You may use MATLAB or Python programming languages to complete this project. Prepare a proposal to describe your idea and suggestions of your solution plan. You'll need few literature papers to support and justify your solution.

Report your work in a technical paper format.

INSTRUCTIONS:
You must complete the following progress tracking:

(A) Complete THREE (3) formative assessments:

i. Practical Assessment 1: Project Proposal (10%)
ii. Practical Assessment 2: Fuzzy Solution Design (10%)
iii. Practical Assessment 3: Agentic Solution Design (10%)

(B) Present your progress on Week 13 (8th-12th June, 2026) during the lab sessions. Each group will be given 20 mins for the presentation.

(C) Present your Project Output on Week 14 (22nd-26th June, 2026) during the lab sessions. Each group will be given 20 mins for the presentation

Your presentation should include:

i. Project Introduction in Brief (3 mins)
ii. Proposed Solutions (5 mins)
iii. Project Findings, i.e. the experimental results, performance evaluation, and results discussion, etc. (10 mins)
iv. Conclusion (2 mins)

(D) Submit the followings in a ZIP file to the mini project submission link in the ULEARN system by 28/6/2026 (Sunday).

i. All your working files
ii. All your primary references
iii. A 6-page technical report (IEEE conference format). Refer to the technical report template provided (**TechnicalPaperSample.docx**).

BAXI3123 FUZZY LOGIC
Group Project Evaluation Form

Project Title:

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Group Members: (i) ______________________________ **(ii)** ______________________________

(iii) ______________________________ **(iv)** ______________________________

(v) ______________________________ **(vi)** ______________________________

(vii) ______________________________

Presentation Date: ____________________ **Presentation Time:** ______________________________

<table>
<tr><th rowspan="2">No.</th><th rowspan="2">Elements</th><th colspan="2">Marks</th></tr>
<tr><th>Obtained Marks</th><th>Full Marks</th></tr>
<tr><td>1.</td><td>Technical Paper
Organisation, Language (1)
Contents (Correct, Compact & Precise) (1)
Result Presentation & Analysis (Justifiable) (3)</td><td></td><td>5%</td></tr>
<tr><td>2.</td><td>Presentation
Slides (Compact & Precise) (1)
Overall Experiment Set-Up (2)
Overall organisation of Presentation (2)</td><td></td><td>5%</td></tr>
<tr><td>3.</td><td>Individual Evaluation
(*including the Q&A during the presentation)
Leadership/Teamwork & Commitment (5)
Quality of Work (5)
Problem Solving (5)
Solution Programs (5)</td><td>(i)
(ii)
(iii)
(iv)
(v)
(vi)</td><td>20%</td></tr>
<tr><td colspan="2" align="right">TOTAL</td><td></td><td>30%</td></tr>
</table>

Overall Comments:

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